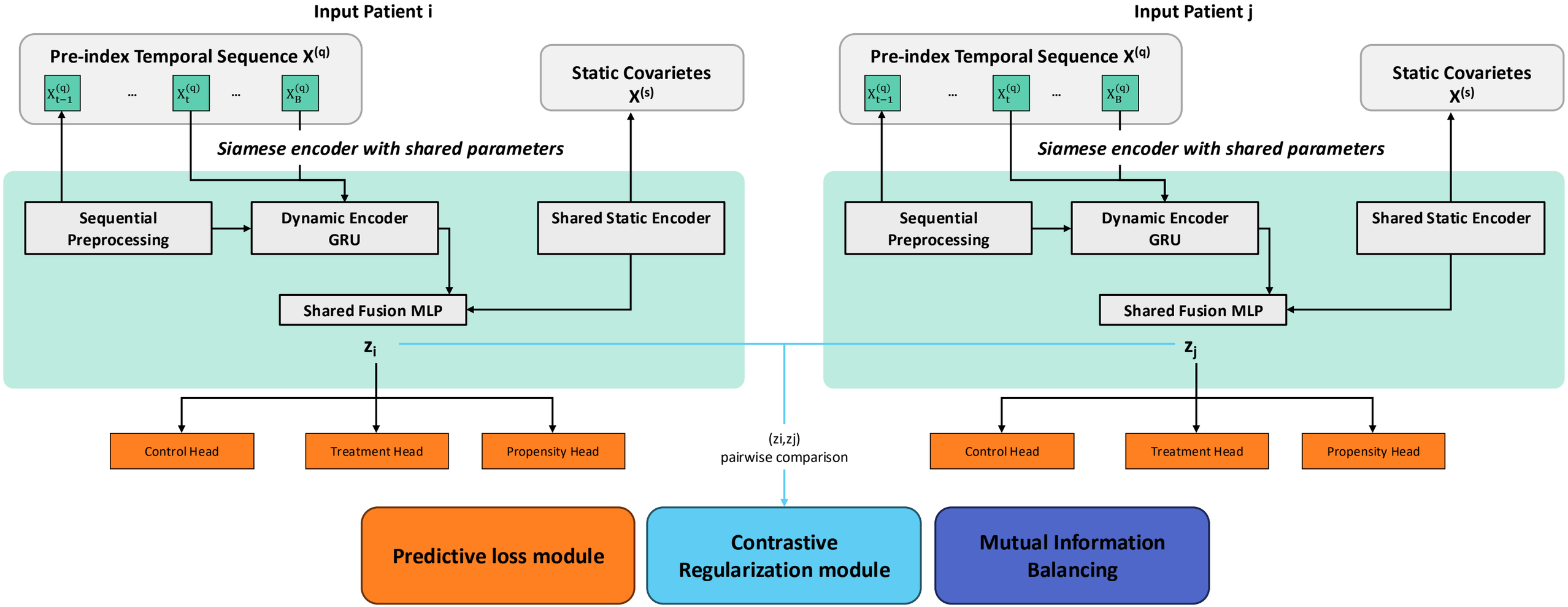
Input Patient i
Input Patient j
Pre-index Temporal Sequence $X^{(q)}$
$X_{t-1}^{(q)}$ ... $X_t^{(q)}$ ... $X_B^{(q)}$
Static Covarietes $X^{(s)}$
Siamese encoder with shared parameters
Sequential Preprocessing
Dynamic Encoder GRU
Shared Static Encoder
Shared Fusion MLP
z_i
z_j
Control Head
Treatment Head
Propensity Head
(zi,zj) pairwise comparison
Predictive loss module
Contrastive Regularization module
Mutual Information Balancing